

$$2.) p(x) = a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0$$

$$p(0) = a_4 0^4 + a_3 0^3 + a_2 0^2 + a_1 0 + a_0 = 1 \rightarrow a_0 = 1 \quad \text{Sikovaná premenná na 1}$$

$$p(1) = a_4 1^4 + a_3 1^3 + a_2 1^2 + a_1 1 + 1 = 2$$

↓

$$a_4 + a_3 + a_2 + a_1 = 1$$

$$p(2) = a_4 2^4 + a_3 2^3 + a_2 2^2 + a_1 2 + 1 = -3$$

↓

$$16a_4 + 8a_3 + 4a_2 + 2a_1 = -4$$

$$\rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 \\ 16 & 8 & 4 & 2 \end{pmatrix} \begin{pmatrix} a_4 \\ a_3 \\ a_2 \\ a_1 \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

$a_0^2 + a_1^2 + a_2^2 + a_3^2 + a_4^2$ je najmenšia možná

$\hookrightarrow$ norma $(a_4, a_3, a_2, a_1)^T$ je najmenšia možná

najmenšia norma:

$$a_{\text{ls}} = A^+ z; \quad A A^+ z = b$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 16 & 8 & 4 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 8 \\ 4 \\ 2 \end{pmatrix} \xrightarrow{z} = \begin{pmatrix} 1 \\ -4 \end{pmatrix} \rightarrow \begin{pmatrix} 4 & 30 \\ 30 & 340 \end{pmatrix} \vec{z} = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 4 & 30 & 1 \\ 30 & 340 & -4 \end{array} \right) \sim \left(\begin{array}{cc|c} 4 & 30 & 1 \\ 15 & 170 & -2 \end{array} \right) \sim \left(\begin{array}{cc|c} 20 & 150 & 5 \\ -15 & -170 & 2 \end{array} \right) \sim \left(\begin{array}{cc|c} 20 & 150 & 5 \\ 5 & -20 & 7 \end{array} \right) \sim \left(\begin{array}{cc|c} 20 & 150 & 5 \\ -20 & 80 & -23 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{cc|c} 4 & 30 & 1 \\ 0 & 230 & -23 \end{array} \right)$$

$$230z_2 = -23$$

$$z_2 = -\frac{23}{230} = -\frac{1}{10}$$

$$4z_1 = 1 + \frac{30}{10} = 1 + 3 = 4$$

$$z_1 = 1$$

$$a_{\text{ls}} = \begin{pmatrix} 1 & 16 \\ 1 & 8 \\ 1 & 4 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -\frac{1}{10} \end{pmatrix} = \begin{pmatrix} 1 - \frac{16}{10} \\ 1 - \frac{8}{10} \\ 1 - \frac{4}{10} \\ 1 - \frac{2}{10} \end{pmatrix} = \begin{pmatrix} \frac{10-16}{10} \\ \frac{10-8}{10} \\ \frac{10-4}{10} \\ \frac{10-2}{10} \end{pmatrix} = \begin{pmatrix} -\frac{6}{10} \\ \frac{2}{10} \\ \frac{6}{10} \\ \frac{8}{10} \end{pmatrix} = \begin{pmatrix} -\frac{3}{5} \\ \frac{1}{5} \\ \frac{3}{5} \\ \frac{4}{5} \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -3 \\ 1 \\ 3 \\ 4 \end{pmatrix}$$

$$p_{\text{ls}}(x) = \frac{1}{5} (-3x^4 + x^3 + 3x^2 + 4x + 5)$$